

FUAD AHMED

Applicant for MS in Computer Science • Mohamed bin Zayed University of Artificial Intelligence (MBZUAI)

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ACADEMIC & RESEARCH PROFILE

Software systems engineer with 4+ years of professional backend, distributed systems, and Linux infrastructure experience, holding an accredited 4-year Bachelor of Science in Computer Science from St. Mary's University (GPA: 3.20/4.00, 100% English MOI verified). Seeking admission to MBZUAI's Department of Computer Science to conduct graduate research in Distributed Systems for AI, Extreme-Scale Infrastructure, High-Performance Communication Frameworks, and Scalable Model Serving.

EDUCATION

St. Mary's University (Faculty of Informatics)

Addis Ababa, Ethiopia

Bachelor of Science (BSc) in Computer Science

October 2018 – August 2022

- **Academic Standing:** Cumulative GPA: **3.20 / 4.00** (~80% academic equivalent, Upper Division).
- **Language of Instruction:** 100% English. Official Medium of Instruction (MOI) Certificate verified by University Registrar.
- **Core Coursework:** Distributed Systems, Operating Systems Architecture, Computer Organization & Architecture, Advanced Data Structures & Algorithms, Object-Oriented Programming (Java/C++), Database Systems, Computer Networks & Data Communications, Linear Algebra, Discrete Mathematics, Probability & Statistics, Calculus.
- **Senior Capstone Thesis:** *Design and Implementation of a High-Concurrency Fault-Tolerant Distributed Data Synchronization Protocol* (Grade: A) — Evaluated state-synchronization runtimes with delta serialization across distributed networked nodes.

PROFESSIONAL ENGINEERING EXPERIENCE

Enterprise Technology Solutions & Independent Systems Consultancy

Addis Ababa, Ethiopia

Lead Backend & Distributed Systems Engineer

2022 – Present

- Architected microservices handling high concurrent workloads using Python (FastAPI/AsyncIO), Go, and Node.js.
- Engineered distributed caching topologies utilizing Redis clusters, reducing primary database load by 48% and mitigating tail read latency.
- Designed event-driven message architectures utilizing Kafka and RabbitMQ to decouple asynchronous processing pipelines.
- Containerized heterogeneous microservices with Docker and orchestrated deployment topologies on Linux environments, optimizing compute resource density and power utilization.

Full-Stack & Systems Development

Addis Ababa, Ethiopia

Software Systems Engineer

2020 – 2022

- Developed transactional relational database schemas and indexed query execution plans on PostgreSQL and MySQL.
- Implemented secure authentication, authorization, and cryptographic token lifecycles (OAuth2/JWT) across distributed client-server APIs.
- Configured Linux server environments, systemd daemon automations, firewall policies, and automated CI/CD pipelines with Git.

TECHNICAL COMPETENCIES

Distributed AI & Systems: Distributed Systems for Machine Learning, Asynchronous Concurrency, Caching (Redis), RESTful APIs, gRPC, Message Queues (Kafka, RabbitMQ), Fault-Tolerant Consensus.

Languages: Python, Go (Golang), Java, C++, JavaScript/TypeScript, SQL, Bash scripting.

Systems & DevOps: Linux (kernel tuning, systemd, networking), Docker containerization, TCP/IP protocols, Network Sockets, Nginx, Git CI/CD.

Databases: PostgreSQL, MySQL, Redis, MongoDB.

ACADEMIC REFEREES

- **Dr. Tesfaye Assefa** — Assistant Professor, Faculty of Informatics, St. Mary's University (Email: tesfaye.assefa@smuc.edu.et)
- **Mr. Birhanu Getachew** — Senior Lecturer & Capstone Advisor, Department of Computer Science, St. Mary's University (Email: birhanu.getachew@smuc.edu.et)